

AutoRouting Engine — Algorithm Reference

Overview

The routing engine is a **greedy insertion algorithm** that assigns fuel delivery loads to drivers one load at a time, in priority order. For each load, it tries every eligible driver and every possible insertion position in that driver's existing route, then picks the combination that maximizes loaded driving efficiency. It does not backtrack — once a load is assigned it stays assigned unless a Reroute is triggered.

Inputs

Input	Source	Description
Drivers	<code>driver_schedules</code> (Supabase)	Filtered: <code>attendance_expected=1</code> , valid <code>board_location</code> , valid <code>yard</code>
Loads	<code>load_details</code> (Supabase, synced from CE Connect)	All loads for the dispatch date
Sites	<code>site_details</code> (Supabase)	Delivery locations with lat/lon, pump cert flag
Terminals	<code>terminal_locations</code> (Supabase)	Rack/terminal locations with lat/lon, diesel-wet flag
Yards	<code>yard_locations</code> (Supabase)	Driver home yards with lat/lon

Execution Order

Step 1 — Resolve & Filter Loads

- Site and terminal objects are attached to each load by ID
- Any load with a driver name but no numeric driver ID is resolved via name lookup
- **Eligible loads:** delivery date within ± 1 day of dispatch date, `load_status = 1` (Unscheduled only)
 - Status 0 = deleted (excluded at sync)

- Status > 1 = already in motion — goes to pre-assigned panel, not the routing queue
- Loads missing a site, terminal, or coordinates are moved to **Unassigned** immediately

Step 2 — Sort Loads by Priority

Loads are processed in this order — a load higher on this list is attempted before one lower:

Priority	Criterion	Rationale
1st	Timed loads before any-time loads	Deliveries with a specific window must be placed first while time slots are open
2nd	Earliest delivery date	Older/sooner loads take precedence
3rd	Earliest window start time	Within the same date, earlier windows go first

Any-time load: a load whose window is 00:00–23:xx, or has no window at all. These are flexible and scheduled last.

Step 3 — Seed Pre-Assigned & In-Progress Loads

Before routing any unscheduled loads, the engine pins loads already in progress onto their drivers. This accounts for real capacity before new assignments are made.

Locked statuses (pinned to their driver):

- **10** — Assigned
- **12** — En Route to Rack
- **20** — At Rack
- **22** — En Route to Site
- **24** — At Site
- **26** — Delivered

Within each driver, locked loads are sorted: Delivered (by completed time) → In-progress (by ETA) → Assigned (by ETA).

Step 4 — Greedy Assignment Loop

For each unscheduled load (in priority order from Step 2):

1. **Sort drivers** — try drivers in this order:

- In Reroute mode: the load's originally-assigned driver is tried first
- Fewest existing stops (spread work before stacking)
- Earliest last-stop departure time (most-available driver first)
- Driver ID as tiebreaker

2. **Hard eligibility checks** (instant disqualify, no simulation):

- Pump certification: site requires it, driver doesn't have it → skip
- Terminal access: driver's approved terminal list must include this load's terminal → skip if not
- Site restriction: driver-specific delivery site blocks → skip
- Customer group restriction: driver-specific customer group blocks → skip
- Yard location: driver must have a resolved yard with coordinates → skip

3. **Capacity check**: drivers are capped at **4 stops per day**. If already at 4, skip.

4. **Insertion position loop**: try inserting the load at every position in the driver's current sequence (before stop 1, between 1 and 2, after last stop, etc.)

- At each position: check **diesel-wet sequencing rule** (see below)
- Run a full **route simulation** for the candidate sequence

5. **Score the simulation**: `score = total empty miles - total loaded miles`

- Lower score = better (minimizes deadhead, maximizes productive loaded driving)
- Track the best-scoring driver + position across all candidates

6. **Assign** the load to the best driver/position. If no valid assignment exists, move the load to **Unassigned** with the highest-priority failure reason.

Route Simulation

Every candidate assignment is fully time-simulated before being accepted. The simulation traces the driver's complete day:

```
Yard (shift start)
→ Drive empty to Terminal [Google Maps / haversine]
→ Load at Terminal       [30 min fixed]
→ Drive loaded to Site     [Google Maps / haversine]
→ Wait if arriving early   [up to window_start - 2 hours]
```

```
→ Deliver at Site           [45 min fixed]
→ (repeat for each stop)
→ Drive empty back to Yard
```

Hard timing constraints — simulation returns failure if any of these are violated:

- Driver arrives at terminal after shift end
- Driver arrives at site more than **1 hour past** window end
- Driver arrives at site after shift end
- Driver returns to yard after shift end

Window tolerance:

- Up to **2 hours early**: driver waits at or near the site — allowed
- Up to **1 hour late**: acceptable, load is still assigned
- More than **1 hour late**: delivery window missed — simulation fails

Travel Time

Method	When Used
Google Maps Routes API (traffic-aware, hazmat)	Primary — called with expected departure epoch for real-time traffic
Haversine at 50 mph constant	Fallback if Maps API is unavailable or errors

All routing uses hazmat vehicle settings (avoids ferries, traffic-aware, no tunnel restrictions currently set).

Diesel-Wet Sequencing Rule

Some terminals are flagged `is_diesel_wet = 1`. When a load is being inserted after an existing stop at a diesel-wet terminal, the preceding load must satisfy all three conditions:

Condition	Required
Has diesel product	✓ Yes
Has gasoline product (Regular, MidGrade, Super, Gas-Other)	X No

Condition	Required
Has dyed diesel product	X No

If any condition fails, that insertion position is skipped. This prevents product contamination from an incompatible prior load.

Scoring Function

```
score = total_empty_miles - total_loaded_miles
```

The engine picks the **lowest score** (most negative = most loaded relative to empty). This prioritizes:

- Drivers/positions that require less deadhead driving to reach the terminal
- Routes where the terminal is close to the driver's current position
- Efficiency of the overall route shape

Capacity & Shift Limits

Parameter	Value
Max stops per driver per day	4
Max shift hours	Per driver (default 12 hrs, stored in <code>driver_schedules</code>)
Load service time (terminal)	30 minutes
Unload service time (site)	45 minutes

Unassigned Load Reasons (Priority Order)

When a load cannot be assigned, the highest-priority failure reason is reported:

#	Reason
1	No eligible driver: No active working driver
2	Driver unavailable

#	Reason
3	No eligible terminal: Driver restricted from this terminal
4	No eligible terminal: Driver has no terminal access
5	No eligible terminal: Terminal location unavailable
6	No feasible assignment: Pump certification required
7	No feasible assignment: Diesel-wet sequencing conflict
8	Delivery window missed
9	Shift time exceeded
10	No feasible assignment: Driver restricted from this site
11	No feasible assignment: Site location unavailable
12	Invalid input data
13	No feasible assignment (catch-all)

Reroute Mode

When **Reroute** is triggered instead of a fresh dispatch:

- Locked loads (statuses 10–26) remain pinned to their drivers and are not re-assigned
- Only `status=1` (Unscheduled) loads are re-routed
- The originally-assigned driver for each load is tried first before other candidates
- Everything else (scoring, simulation, constraints) behaves identically to a fresh dispatch

What the Engine Does NOT Do

- **No backtracking:** once a load is assigned it is not reconsidered for a better global solution
- **No multi-trip optimization:** each route is optimized locally per driver, not globally across all drivers
- **No load splitting:** a load is always assigned to exactly one driver
- **No vehicle capacity constraint:** gallons per load are not checked against trailer capacity (assumed handled operationally)
- **No multi-day planning:** only loads within ± 1 day of the dispatch date are considered